

Battlemon hardening and native validation

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Base: `caa49b075253d21d05160f04d4e525ffe5b634c0`

Branch: `battlemon`

Implemented policy

- Pet movement and autonomous behavior remain native-client owned. The server does not invent a second AI loop.
- The complete client EXP table drives persisted/displayed progression from level 1 through 250. Static executable tracing separates that value from the native AI-profile level: all twelve recovered pet actor-construction paths pass AI level 1, while `AI_PetA` - `AI_PetK` independently contain difficulty profiles 1-13. Level 14 is therefore admitted and EXP is not capped at 4,807.
- Every relay session containing an owned pet installs an actor-ownership policy. Unknown and malformed inner relay packets are dropped rather than forwarded. Sessions without owned pets retain the existing relay behavior.
- Skill and spell reports require a live, authorized actor and a gameplay endpoint. Non-gameplay endpoints and spectators cannot submit combat. A nonzero spell hit must consume a recent matching skill authorization; each authorized target can be consumed once. Server-scheduled DoT ticks do not re-enter this client-report gate.
- Enhanced sessions accept one point report after a relay-observed serve. The atomic gate suppresses duplicate reports from separate authorized endpoints; the packet decoder independently rejects an immediate repeated packet serial. This is live-session duplicate suppression, not a durable replay ledger: the native header serial wraps after 60 packets and `CMSGPoint` contains no rally nonce that could distinguish a later byte-identical legitimate result.
- Enhanced-match reward persistence and the durable result claim share one serializable transaction. A random per-session UUID identifies concurrent or repeated completion calls for that live session; an independent claim token distinguishes inserts from duplicate-key no-op updates without relying on connector affected-row behavior. Transaction failure rolls back the claim and persistence and closes participating clients so rolled-back in-memory mutations cannot remain usable. This provides atomic duplicate suppression, not durable delivery after a process restart or a failed completion.
- Lifecycle is an explicit compatibility policy: lazy whole-day decay under a pessimistic pet lock, hunger -1/day and energy -4/day. Expiry or zero hunger marks a pet dead; zero energy blocks gameplay but does not mark it dead. A null lifecycle cursor initializes without retroactive decay.
- `PET_ITEM` effects implemented: indices 1-4 (+1 stat), 5-8 (+2), 9-12 (+5), 13 (+1 validity day), 14 (+5 `lifeMax`), 16-19 (hunger), 20-22 (energy), and 23 (+50 hunger and energy), with species caps and a stat cap of 127. Index 15 remains unknown and fails without consumption.

- Dedicated Battlemon deliberately remains the evidenced topology: two owners at 0/1 and one owned pet each at 2/3. Guardian `Allow Battlemon` remains an all-active-owner rule. Unsupported dedicated topologies and spectators fail admission rather than being extrapolated.
- Reconnect is not synthesized. Disconnecting a gameplay owner aborts and cleans the enhanced match, resets the room, removes relay authorization, and returns remaining clients to the room. A fresh match is required. A transient empty relay connection list does not remove actor authorization; only the explicit game-server cleanup message does, so reconnect cannot fail open.

Automated verification

- Full Maven reactor regression from the final source tree: 130 game-server tests, 9 relay-server tests, and all other module suites passed.
- SQL integration checks against MySQL 8.0.23 prove result-claim rollback and concurrent duplicate behavior: the second transaction observes the first claim token and cannot claim the same result.
- A transactional MySQL 8.0.23 check proves `Pet.level` stores and reads 250 as `TINYINT UNSIGNED`; the packet regression proves that Java `Integer(250)` is emitted as the single wire byte `FA`.
- The final focused/full rerun and package logs are stored under `.amp/tmp/battlemon-hardening-e2e/logs/`.

Native observations

- Two 1280×800 recordings were captured at 60/1 FPS for the earlier level-boundary experiment: 300 seconds and 18,000 frames per client. The level-14 pet and 4,808 EXP rendered in the unmodified client. That run's admission rejection was the former server policy, not evidence of a native level-14 incompatibility; static actor-construction evidence has superseded that policy. The level-14 recording has SHA-256 `c1732bf5c175e612df322606764741bc7971847b818a37e56391a999f37fb2cb`.
- A fresh client run against the final packaged emulator rendered the level-14 fixture at 4,808 cumulative EXP and the level-250 fixture at 1,408,515 cumulative EXP. Both pets could be selected without disconnecting. This final boundary run is supported by native screenshots and packet regressions. Its 60 FPS capture ended at the login screen before the later UI automation and therefore is not claimed as evidence of the pet dialogs.
- A further 1280×800, 60 FPS, six-minute native recording covers the `PET_ITEM` setup path. The unmodified client listed the fixture items, rejected feeding with no selected Battlemon without a database mutation, then emitted `CMSGPickupPet` for PotekoTest and accepted `SMSGPickupPet result=0`. Returning from Housing reset the Wine client before a positive item mutation, so this run does not claim native validation of every implemented effect. The item fixture was restored exactly afterward.

- The final packaged JARs were deployed together in the emulator container. Two isolated native clients then authenticated (`SMSGLogin result=0`) and entered the game server (`SMSGLoginData result=0`); both authoritative player rows were online on `GAME_SERVER` . A post-deployment 1280×800, 60 FPS, 120-second recording contains 7,200 frames. This is a release smoke test, not additional proof of the PET_ITEM effects or match edge cases.

Raw local evidence is retained under `.amp/tmp/battlemon-hardening-e2e/` :

- `video/final4-level1-control-b.mp4` — 1280×800, 60 FPS, 18,000 frames.
- `video/final4-level14-subject-c.mp4` — 1280×800, 60 FPS, 18,000 frames.
- `video/final-all-pet-items-60fps.mp4` — 1280×800, 60 FPS, 21,600 frames, SHA-256 `cb45dfca2405d3cd27773eddf09beea5dbc83ed5894ed46cf379cb855af4f286` .
- `video/final-postdeploy-native-60fps.mp4` — 1280×800, 60 FPS, 7,200 frames, SHA-256 `bc9995779daebe41e5cb630aa3eb30f50d5f7685bacc36d78ca505e9ecec0240` .
- `db/post-all-pet-items-restore.txt` — post-run fixture and account-state restoration proof.
- `logs/final-source-full-maven-test.log` and `logs/final-postreview-package.log` — regression and release packaging from the exact final source tree.

Non-claims

- Hunger/energy rates and revive expiry remain compatibility decisions. The 127 stat boundary is native-observed for the current signed-byte path. The former level-13 admission rule has been removed because native AI difficulty and displayed pet progression are separate values.
- Result de-duplication does not survive reconstruction of a match into a new `GameSession` , and failed completions are not durably queued for retry.
- Point replay suppression is bounded by the live rally gate and packet serial validation. Restart-safe or wrap-safe replay protection requires a protocol rally identifier that the retail `CMSGPoint` packet does not carry.
- The native PET_ITEM run proves enumeration, selection preconditions, and pet pickup, but not every positive item effect; those effects currently have automated service/packet coverage.
- No new meaning is assigned to unknown controller direction/command fields or unknown relay payloads.
- Index 15, retail ranking/matchmaking formulas, alternate dedicated topologies, mid-match reconnection, and historical balance are not claimed.